BCNF :

Student

Sid	Sname	Pname
01	Raj	Ram
02	Rio	Ram
03	Raja	Ram

Student

sid	Sname	Pid
01	Raj	01
02	Rio	01
03	Raja	02

Professor

Pid	Pname
01	Ram
02	Rami

WINDOW FUNCTION

A window function performs a calculation across a set of related rows while keeping each individual row in the result

OVER () clause

The OVER() clause tells SQL how to apply a window function to rows

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```
SELECT ename, salary
    RANK() OVER ( ORDER BY salary DESC)
AS salary - rank
FROM employee;
```

OVER() - defines the window for the calculation

Types of ranking function

1. RANK()
2. DENSE_RANK()
3. ROW_NUMBER()

1. **RANK()** : assigns a rank to each row, If two employee have the same salary, they receive the same rank, and the next rank is skipped.

```
SyntEx:  SELECT ename, salary
             RANK() OVER (ORDER BY salary DESC)
         AS salary-rank
         FROM employee;
```

O/p:

Employee	salary	salary-Rank (GAP)
Arun	50000	1
Priya	46000	2
Rahul	40000	2
Divya	30000	4

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2. DENSE_RANK () : This also gives the same rank to employees with equal salaries, but it does not skip the next rank.

```
Ex: SELECT ename, salary
      DENSE_RANK () OVER (ORDER BY salary DESC)
    AS salary_rank
    FROM employee;
```

o/p:

Employee	salary	salary_rank. (NO GAP)
Arun	50000	1
Priya	40000	2
Rahul	40000	2
Divya	30000	3

3. ROW_NUMBER () : assigns a unique no. to every row, even when salaries are equal

```
Ex: SELECT ename, salary
      ROW_NUMBER() OVER (ORDER BY salary DESC,
        ename)
    AS row_num
    FROM employee;
```

Here, ename is an additional sorting column

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to make the ~~row~~ order of employees with equal salaries predictable.

o/p :

Employee	salary	Row_Num
Arun	50000	1
Priya	40000	2
Rahul	40000	3
Divya	30000	4

Functions	Same Salary gets same rank?	Skips ranks?
RANK()	yes	yes
DENSE_RANK()	yes	NO
ROW_NUMBER()	NO	NO

PARTITION BY : This divides rows into groups and perform a window calculation separately within each group. Unlike GROUP BY, it keeps all original row.

Ex: SELECT ename, department, salary,
RANK() OVER(PARTITION BY department ORDER BY salary DESC)
AS salary_rank
FROM employee;

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Imagine the employee table has a department column

Employee	Department	Salary
Arun	IT	50000
Priya	IT	40000
Rahul	HR	45000
Divya	HR	30000
Kiran	Sales	35000

Suppose we want to rank employees within each department, rather than across the entire company

O/P:

Employee	Department	Salary	Rank
Arun	IT	50000	1
Priya	IT	40000	2
Rahul	HR	45000	1
Divya	HR	30000	2
Kiran	Sales	35000	1

Rahul receives rank1 even though arun earns more. why? Because PARTITION BY department starts a separate ranking for each department without PARTITION BY, all employees would be ranked together.

Page No.: ..

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GROUP BY	PARTITION BY
• combines rows into groups	• Divides rows into calculation groups
• Usually returns one row per group when used with aggregate functions	• Keeps individual rows
• commonly used with aggregate functions	• used with window functions.

GROUP BY : This returns one row per department

PARTITION BY : This returns every employee, along with their department.

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